

Post Pregnancy Diabetes Predictor

1. INTRODUCTION

India is deemed as the world's capital of diabetes. The diabetic population in the country is close to hitting the alarming mark of 69.9 million by 2025 and 80 million by 2030. This denotes that the developing country is expected to witness an increase of 266%. Diabetes is a chronic disease characterised by high blood sugar. It may cause many complicated disease like stroke, kidney failure, heart attack, etc. Diabetes is said to be the leading disease in India, with many new cases being reported each year. It is a long-lasting disease which occurs due to high blood glucose and leads to various health problems including heart diseases, kidney damage, strokes, and vision issues. Post-pregnancy, women tend to have a greater risk of getting diabetes due to changes in their hormones and lifestyle. Hence, there is a need to develop an early prediction and detection model of post-pregnancy diabetes for the prevention of such cases. In today's era, Machine Learning (ML) approaches have become quite efficient and popular for predicting various diseases. The proposed project is about implementing a ML approach to predict post-pregnancy diabetes with the use of health parameters and medical characteristics of patients. The main objective of this study is to develop a machine learning (ML)-based model for predicting diabetes post pregnancy. Logistic regression (LR) is used to identify the risk factors for diabetes. In this research, diabetes is predicted using significant attributes, and the relationship of the differing attributes is also characterised.

2. PROBLEM STATEMENT:

The traditional methods of identifying the presence of diabetes in patients generally involve multiple medical checkups. Moreover, many currently employed prediction models are not very accurate because of their unreliable datasets and lack of feature analysis. In order to develop a solution to this problem, we intend to create a smart and efficient prediction system that will utilize Machine Learning and healthcare data analysis for predicting the probability of developing diabetes post-pregnancy.

3. OBJECTIVE:

So as far what we have discussed is, what our project actually do. The objective of this research is to make use of significant features, design a prediction algorithm using Machine learning and find the optimal classifier to give the closest result comparing to clinical outcomes. The proposed method aims to focus on selecting the attributes that ail in early detection of Diabetes Post Pregnancy using Predictive analysis. The result shows the decision tree algorithm and the Logistic Regression has the highest specificity of 70.20% and 82.20%, respectively holds best for the analysis of diabetic data. The research also generalizes the selection of optimal features from dataset to improve the classification accuracy. The primary goals of this project are as follows:

- Develop a Machine Learning algorithm to predict diabetes after pregnancy.
- Investigate critical medical variables that influence diabetes.
- Determine key risk factors through Logistic Regression.
- Enhance early detection and health care decisions.
- Design a straightforward, effective, and accurate predictive method.

4. SCOPE OF THE PROJECT:

Scope of this project is the prediction of the likelihood of diabetes with the help of machine learning algorithms and medical data of the patients. This project will be helpful for doctors and other healthcare workers to find out risky patients, hence playing an important role in preventive healthcare. Also, this project will pave the way for further improvements like real-time health tracking, web and mobile health care apps, and incorporation of advanced non-invasive diabetes detectors.

5. LITERATURE SURVEY

There have been several prediction systems for diabetes created using Machine Learning techniques like Decision Trees, Support Vector Machine (SVM), Random Forest, and Logistic Regression. But most of the prediction systems lack accuracy and have inconsistent datasets. The proposed system will improve prediction efficiency by implementing advanced Machine Learning techniques and improved data pre-processing techniques. There are many such “Diabetes Predictor” models, but our model is more

accurate as compared to theirs. Today if a sanitiser kills 99.99% of germs then people are more concerned about the rest of 0.01% germs so how will they be satisfied with “70% or less” accuracy.

6. METHODOLOGY:

The first step involved the collection of datasets of healthcare information regarding diabetes. Then the collected dataset was preprocessed in order to get rid of any inconsistency, missing values and duplicate entries. After performing preprocessing on the dataset, important medical attributes associated with diabetes were selected. Logistic Regression machine learning algorithm is employed to train the ML model. Then the ML model analyzes the relationship between health attributes and identifies whether there is a possibility of post-pregnancy diabetes. The performance of the system is evaluated with respect to the prediction accuracy.

7. TECHSTACK:

Python is chosen for the implementation of the project because it has simple syntax and easy to use with various machine learning libraries. Such libraries as NumPy, Pandas, Sklearn, and Matplotlib are utilized for various tasks including data manipulation and visual representation. Platforms such as Jupyter notebook and Google Colab can be used for implementing this project efficiently.

8. SYSTEM ARCHITECTURE:

The system architecture consists of multiple stages beginning with patient data collection. The collected data undergoes preprocessing and feature selection to improve data quality and identify important attributes. The processed data is then provided to the Machine Learning model for training and prediction. Based on the analysis, the system generates a prediction result indicating the likelihood of post-pregnancy diabetes.

9. PROPOSED SYSTEM:

This system has various benefits in the health care sector. The first strength is that the system will aid in identifying patients at the early stage of developing diabetes, thus

helping them take preventive measures. Secondly, this method diagnoses the disease fast and accurately as compared to traditional diagnostic approaches. Furthermore, the system assists healthcare professionals in decision-making and lowers the healthcare burden through automated prediction approaches. Finally, it is cost-effective and easy to use.

10. LIMITATIONS:

Every system has its weaknesses. In this case, the main limitation is the accuracy of the prediction model. This is because the quality and consistency of the data used in building the model can determine its efficiency. In this case, there are some inconsistencies in the dataset obtained, hence affecting the prediction accuracy. However, efforts are currently being put into improving data consistency to enhance the model's accuracy and dependability.

11. PREVIOUS SYSTEM:

There are several methods that have been created for diabetes predictions before, but most of them lack high prediction efficiency and accuracy. In the field of healthcare, however, the slightest improvement in prediction efficiency may be crucial because it affects the lives of patients. A lot of classical models are not able to offer highly efficient predictions due to insufficient data and inefficient algorithms used. The suggested method aims at improving the prediction efficiency by enhancing the Machine Learning process and increasing the efficiency of analyzing healthcare data.

12. CONCLUSION:

In summary, the “Post Pregnancy Diabetes Predictor” is a project that illustrates the implementation of Machine Learning techniques in predicting risks of developing diabetes in advance. The project helps identify high-risk individuals by means of medical information analysis. In general, it is an effective, reliable, and user-friendly solution which can help healthcare specialists offer better treatment and prevention options to patients. In the future, it will be possible to improve prediction accuracy, integrate advanced non-invasive methods of diabetes detection, and reduce reliance on painful conventional blood sugar tests.

13. FUTURE SCOPE:

In order to enhance the project, it is necessary to implement advanced Machine Learning algorithms in order to increase the accuracy rate. It will be possible to create applications,

both mobile and web-based ones, for increased convenience. Real time monitoring systems for patients' health condition analysis will also be considered. Larger amounts of healthcare databases can be used in order to improve system performance. In addition, it will be possible to introduce advanced non-invasive diabetes detection methods.

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